



Instructor: Date:

Learner: Regional Branch:

Program: Center:

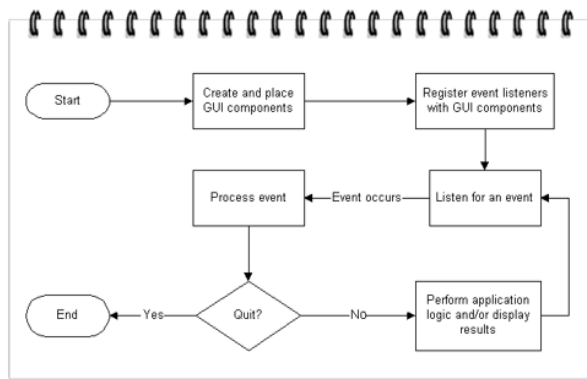
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Let's Start!

The Use of Hierarchy in Data Flow Diagrams.



Learning Outcome:

Describe common activities at work using short sentences in familiar spoken and written situations.

Interpret the requirements report to determine the technological needs in managing information, based on rules and protocols established by the company.

Can Do Statements:

- I can explain how the hierarchy works in data flow diagrams.
- I can describe different levels of hierarchy in data flow diagrams.
- I can talk about detailed facts related to data flow diagrams.

Ask your partner:

- Where can you find hierarchy structures? Explain.
- Why do you think is hierarchy important?



Let's Explore!

Taking into account* Hierarchy in Data Flows Diagrams.

*"Take into account" is an expression that means to consider.



Instructions: Read the text below.

Instructor: Before, people didn't organize information in hierarchy diagrams, but analyzers became aware of the advantages of representing processes in a chain of command. For instance, in a DFD, hierarchy works as a way to organize better the information and the communication. When you analyze a diagram, you see that the various levels are representing the degrees of details. It also shows the whole or a specific area of interest in the system. So, the users find facts easily. This is also very useful when they are making decisions or when they are doing any agreement.

There are different levels of hierarchy in data flow diagrams. I created one here. Look! In level 1, I drew the entire system; we call this the top-level. In level 2, I organized the expansion of processes; actually, this is representing just a part of the whole process. On the other hand, I didn't decompose these other processes because they are simple. In that case, they go in the bottom-level process. Remember, unless you have a complex process, you draw a new level.

There are also some detailed facts you need to consider. First, unless you have more than 7 or 9 processes in a DFD, you need to include another level. In that case, it's necessary to group the other processes and to represent them smaller with just one process in a diagram that encompasses all. With that new grouped process, you organize another lower level which shows all the constituents. Unless you have more processes, you decompose the DFD, which you break down into subprocesses and it becomes another lower level. Finally, this is not a tendency, but unless a client doesn't understand all the information, you need to have the same hierarchy in the flow diagram.



Let's Practice!

Hierarchy: A Classification of Levels and Processes.

Instructions: Classify the sentences by writing the number of the sentence in the correct category.

1. When you analyze a diagram, you see that the various levels are representing the degrees of details.
2. You have a complex process, you draw a new level.
3. These data flow diagrams showed different levels of hierarchy.
4. Before, people didn't organize information in hierarchy diagrams.
5. You have more than 7 or 9 processes in a DFD, you need to include another level.
6. You have more processes, you decompose the DFD.

Sentences you need to add "unless" at the beginning	Sentences you don't need to add "unless"



Let's Practice!

I'm going to show you how hierarchy works in data flow diagrams.

Instructions: Find the words in the word search below.

1. I started with level 1, which is called...
2. I organized this type of processes. In level 2, this is representing just a part of the whole process.
3. This type of diagrams are designed to organize.
4. In a DFD, it works as a way to organize better the information and the communication.
5. Unless you have that kind of process, you draw a new level.

I A G B C O M P L E X I E Q J
J D H R J L H P V U A F Z J U
D C I J F U D W U W A T Y L O
Y F E W N Q R Y X Q S O L F E
T C R Q Y Z P A K W C P U L F
E P A Z V O K B B P H L R Z L
F Z R I X C C Q E X Z E Y J O
C J C X C X A Y T K G V B E W
P G H J I P R U W W F E N X U
M Y Y S H M L P L T B L M W P
A F F F S T I B T O U A D W R
N U G W Q A S H M D B O B Q T
L A W N F Q Z U H O E Z D Q W
Z E X P A N S I O N I E F P I
N P F S J Z X Q W B Y B H H G





Extension Activity

Hierarchy in Data Flow Diagrams.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Read some information about levels of hierarchy in DFD.
- In a poster, design and draw a data flow diagram describing hierarchy denotation.
- Present your work to your classmates and check pronunciation with your teacher.

Self-Assessment

Yes

No

- I explain how hierarchy works in DFD.
- I describe different levels of hierarchy in DFD.
- I talk about detail facts related to data flow diagrams.

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Target Words

- encompass** (verb): to have or include all the parts in just one thing
e.g. In that case, it's necessary to group the other processes and to represent them smaller with just one process in a diagram that encompasses all.
- break down** (phrasal verb): to divide into parts
e.g. Unless you have more processes, you decompose the DFD, which you break down into sub processes and it becomes another lower level.
- complex** (adjective): consisting of many parts
e.g. Remember, unless you have a complex process, you draw a new level.

- bottom** (noun): the lowest part of something
e.g. On the other hand, I didn't decompose these other processes because they are simple. In that case, they go in the bottom-level process.
- decompose** (verb): to separate into different parts
e.g. Unless you have more processes, you decompose the DFD, which you break down into sub processes and it becomes another lower level.
- degree** (noun): the steps in a process
e.g. When you analyze a diagram, you see that the various levels are representing the degrees of details.



Learning Points

Zero Conditional With Unless In Affirmative.

You use the **zero conditional** to talk about a situation that is real and depends on one condition. You can use **unless**, which is also a conditional word similar to "if", with this structure.

You use the **zero conditional** to talk about rules, routines, general truths, scientific facts, and cause and effect situations.

This structure has two parts: the first explains a condition and the second explains the result, consequence, or direct relation to the first condition. Remember, both parts need to be in present simple and they always refer to true situations. You must separate the sentences with a comma.

For **zero conditional**, these are the structures you can use.

If + Present Simple + , + Present Simple

Example:

- **If** a diagram has a level, you use a basic graphic.

When the main clause comes first, you don't need a comma.

Present Simple + If + Present Simple

Example:

- You use a basic graphic **if** a diagram has a level.

Unless is also a conditional word similar to **if**. When you use **unless**, you need a main clause and a subordinate clause. If you use **unless** before the main clause, remember to add a comma as well.

Example:

- **Unless** you consider the levels, you can create a diagram.
- You can send the information later **unless** you have time now.

Examples

- Remember, **unless** you have a complex process, you draw a new level.
- First, **unless** you have more than 7 or 9 processes in a DFD, you need to include another level.
- **Unless** you have more processes, you decompose the DFD which you break down into sub processes and it becomes another lower level.
- Finally, this is not a tendency. **Unless** a client doesn't understand all the information, you need to have the same hierarchy in the flow diagram.





Learning Points

What does hierarchy mean?

Hierarchy refers to a scheme which organizes things or people according to their importance.

The hierarchy is the organization of several items or elements in a structure or system which are subordinated according to its values. The items can be located at the top, bottom, or middle of the structure depending on the inferiority and the superiority they represent. Likewise, the items can be depicted either vertically or diagonally.

The word hierarchy is also associated to power; those elements which are at the top, have the power to control the rest of the structure. In our society it is possible to easily identify different organizations with this structure. For instance, the church hierarchy, the military hierarchy, the bureaucratic hierarchy, social hierarchy, and the corporate hierarchy.

In programming, hierarchy diagrams are very important; specifically, data flow diagrams are represented in a chain of command scheme which is set to organize better the information and the communication, representing the degrees of details through several levels.